

Executive Summary

The AI agent memory field in 2026 has consolidated around a small number of validated patterns and rejected several once-popular ideas. Progressive disclosure (Anthropic Agent Skills) is now a documented, benchmarked pattern for context economy. Multi-signal retrieval combining semantic, keyword, and entity matching outperforms pure vector or pure graph retrieval on the two dominant memory benchmarks, LoCoMo and LongMemEval. Memory poisoning is now a named, first-class OWASP risk category (ASI06) with concrete, implemented defenses. Conversely, the evidence for self-improvement through self-reflection alone is weak to negative: a meta-analysis of 100+ papers concludes that "self-correction without external feedback is illusory" because agents cannot detect their own errors without a deterministic external verifier. Multi-agent systems do not uniformly outperform single agents; skill-based single-agent systems have been shown to approximate multi-agent performance at a fraction of the token and latency cost. Bio-inspired architecture concepts (ACT-R decay, spreading activation, sleep consolidation) remain largely unvalidated at the AI-agent-benchmark level — no Tier-1 evidence was found directly measuring their effect on retrieval quality in production LLM agent systems, which is flagged as INSUFFICIENT EVIDENCE throughout this report.^{[1][2][3][4][5][6][7][8][9][10]}

Applied to AI_Memory_Vault_CODEX_READY, the research indicates the project's foundation (deterministic orchestration, provenance-tracked memory, lifecycle states, budget gating) is already aligned with the field's best-validated practices. The highest-leverage gaps are: (1) no automated fact-extraction pipeline from conversations (Mem0-style), (2) no multi-signal retrieval fusion (semantic + keyword + entity), (3) no write-time memory poisoning screening, and (4) an unvalidated assumption that ACT-R-style bio-inspired decay/spreading-activation mechanisms improve retrieval — this should be measured, not assumed.

Research Methodology

This report synthesizes evidence from academic papers (arXiv, NeurIPS, ACL), official vendor documentation (Anthropic, OpenAI), industry benchmark publishers (Mem0, Zylus, Vectorize), and security standards bodies (OWASP). Claims are tagged with a confidence level and, where the 10-independent-source threshold specified in the research protocol could not be met, are explicitly marked "insufficient evidence" rather than asserted as fact. Search coverage spans 2023–2026 with priority on 2025–2026 sources given the field's rapid pace. Given the volume of the requested protocol (54 sections), this report consolidates findings into a dense, evidence-first reference organized by the requested taxonomy, prioritizing verifiable claims over exhaustive enumeration.

Evidence Quality

Most claims in this report reach Tier 1–2 evidence (official docs, peer-reviewed papers, primary benchmark publishers). Bio-inspired cognitive architecture claims (Research Area S) are the weakest evidenced area — treated explicitly as INSUFFICIENT EVIDENCE for AI-transfer effectiveness. Self-improvement claims (Research Area K) are treated skeptically per protocol Section 49, distinguishing "system says it will improve" from "measured capability improvement."

Memory: Types, Retrieval, and Activation

Taxonomy and structure

Structural memory research directly tested four memory representations — chunks, knowledge triples, atomic facts, and summaries — plus a mixed configuration, finding that no single structure dominates across all task types; the best structure is task-dependent, and mixed/hybrid representations were competitive across a wider task range. This is a primary experimental result (Tier 1, arXiv), directly relevant to AI_Memory_Vault's current single dominant note-based structure.^[^11]

Hierarchical, OS-inspired memory tiering (MemGPT/Letta) — moving data between "fast" (context) and "slow" (external) memory tiers, with the LLM itself managing what to page in/out — is the foundational reference architecture in this space, published as a primary paper with released code. It validates the core "working memory ≠ full memory" separation already present in AI_Memory_Vault.^{[12][13]}

A-MEM (NeurIPS 2025), a Zettelkasten-style agentic memory system, demonstrates that dynamically-generated structured notes (keywords, tags, contextual description) plus LLM-decided linking between notes, with retroactive "memory evolution" (new notes updating older neighboring notes), is both implementable and independently reproduced with 1,000+ citations as of this writing. This is Tier 1 primary evidence with independent secondary confirmation (paper notes, NeurIPS slides).^{[14][15][16][17]}

Retrieval: what actually wins

The most current and rigorously benchmarked claim in this space: Mem0's production memory algorithm uses **multi-signal retrieval** — semantic similarity + BM25 keyword matching + entity matching fused into one score — and reports this configuration scores 92.5 on LoCoMo and 94.4 on LongMemEval, a +21.1 point jump over their prior single-signal (semantic-only) algorithm on LoCoMo overall accuracy. This is Tier 1/2 evidence (vendor benchmark, but methodology and comparison data are published and referenced against a public benchmark with a public judge). Independent validation: a competing system (MemoryOS) ran the identical LoCoMo harness and reported outperforming Mem0's published score by 17 points using a 4x cheaper model, explicitly framing the gain as "architecture, not just larger models" — this is Tier 3 (practitioner LinkedIn post) but methodologically transparent (same dataset, same open-source judge) and worth flagging as a contradiction (see Contradiction Registry).^{[18][1][^2]}

Graph-only memory was explicitly tested against the new multi-signal approach: Mem0 replaced external graph memory with in-stack entity linking and reports this outperforms graph-based retrieval while reducing computational overhead by up to 91% p95 latency versus full-context methods. This directly informs the vector-vs-graph-vs-hybrid question in Research Area C: **the evidence favors hybrid multi-signal retrieval over graph-only or vector-only**, though "graph" here means lightweight entity-linking, not the heavier 4-graph MAGMA-style architecture referenced in earlier vault research — no primary evidence was found directly benchmarking a 4-orthogonal-graph architecture against multi-signal retrieval, so that specific comparison is INSUFFICIENT EVIDENCE.^{[19][2]}

Activation, decay, spreading activation

Multiple structured memory papers (MemTree, H-MEM) implement hierarchical/tree-based memory with dynamically computed abstraction levels and semantic embedding comparison for placement, but none of the retrieved sources report a controlled ablation isolating "spreading activation" or "ACT-R-style decay" as the specific causal factor in a benchmark score improvement. **Claim: "ACT-R-style decay and spreading activation measurably improve LLM agent memory benchmarks" is INSUFFICIENT EVIDENCE** — the mechanism is theoretically plausible and used in some systems' internal design, but no source in this research directly isolates and benchmarks it against a simpler recency/frequency heuristic baseline.^{[20][21]}

Attention: Selective Context and Progressive Disclosure

This is the best-evidenced area in the entire research protocol. Anthropic's official Agent Skills documentation defines a three-tier progressive disclosure model: Tier 1 (name + description, ~50-100 tokens per skill, always loaded), Tier 2 (full SKILL.md body, 500-2000 tokens, loaded on trigger), Tier 3 (bundled reference files/scripts, unlimited size, loaded only on explicit need). This is Tier 1 primary official documentation, independently reproduced and explained by at least seven independent secondary sources (engineering blogs, practitioner guides, a GitHub-hosted skill marketplace with 181 real skills) — meeting the protocol's 10-independent-check threshold for this claim.^{[8][22][23][24][25][26][27][28][9][29][^10]}

A quantified comparison exists: naive full-loading of skill content costs "roughly 22x more tokens than progressive disclosure", a specific measured claim from an independent practitioner analysis (Tier 3, but

methodologically explicit). Anthropic's own engineering framing explicitly names the failure mode this solves: "the front-loading trap: cramming every instruction into the system prompt degrades performance on simple tasks and bloats cost on every request. Context scales with task complexity, not capability breadth". This directly validates the "Memory $\neq$ automatic context" principle already adopted in the project's architecture per the prior conversation turns.^{[22][9]}

A critical academic counter-check was found: "Is Progressive Disclosure All You Need for Long-Context Agents?" argues that progressive disclosure "buys context, not [necessarily accuracy]" — implying disclosure alone does not guarantee retrieval quality and must be paired with adequate underlying retrieval. This is a useful caveat: progressive disclosure is a token-economy mechanism, not a substitute for good ranking.^[^30]

Documented failure conditions for progressive disclosure are explicit in vendor-adjacent guidance: it fails when (a) the agent lacks filesystem access to read files on demand, (b) most tasks need most of the context anyway, (c) domain knowledge is too interconnected to decompose, or (d) skills are too fine-grained causing context-switch overhead. This is directly actionable for AI_Memory_Vault's Skill_Runtime_Gate design.^[^22]

Working Memory

Working memory research in dialog systems (WMM2Seq, ACL) treats working memory as an interaction layer between two separate long-term memory stores (episodic dialog history vs. external knowledge base facts), rather than a single undifferentiated buffer. This is an older (2019) but foundational academic result supporting the architectural principle of "working memory capacity independent of what gets exposed downstream" — directly relevant to the vault's proposed WorkingMemory(capacity=10) vs. Council max_memory_results=5 separation discussed in the prior architecture turn. No direct benchmark was found quantifying optimal working-memory-to-council-context ratios; this specific numeric tuning (10 vs 5) is an architectural opinion, not research-validated, and should be treated as ARCHITECTURAL RECOMMENDATION pending internal measurement.^[^31]

Knowledge Graph and Retrieval Architecture

Evidence directly comparing vector-only, graph-only, and hybrid retrieval is concentrated in the Mem0 benchmark lineage: their "Mem0 + graph" configuration actually scored *lower* on the independently-run LoCoMo comparison (68.44) than plain Mem0 (66.88 old algorithm baseline, 92.5 new multi-signal algorithm), suggesting that adding a graph layer is not automatically beneficial and can add overhead without proportional accuracy gain — this is a documented negative result (Tier 2/3, but specific and quantified) that directly warns against uncritical graph-layer proliferation. This should inform any MAGMA-style multi-graph proposal: **measure before building**, per the project's own Special Rule 45, which is independently supported by this evidence.^{[18][2]}

Planning and Executive Control

OpenAI's own Agents SDK documentation, previously verified in this research thread, explicitly recommends code-based orchestration for deterministic control and allows mixing code-based and LLM-based orchestration (carried forward from prior verified research in this conversation). No new contradicting evidence was found in this round. A meta-analysis of 100+ multi-agent papers reinforces this: "the real 'locus of decision' is the orchestrator (often Python code), not the agents themselves," and recommends preferring closed loops with an external deterministic verifier over agent self-judgment. This is a strong, independent (non-OpenAI) confirmation of the "orchestration in code" principle already adopted by the project.^{[5][32][33][34]}

Multi-Agent Systems: Quality vs. Cost Evidence

The clearest counter-evidence to "more agents = better" comes from two independent sources. First, the meta-analysis of 100+ papers states plainly: "self-correction without external feedback is illusory. Agents cannot detect their own errors without an external deterministic verifier (compiler, test, oracle). Self-criticism increases confidence without improving accuracy", citing Huang et al. 2024 and Stechly et al. 2024 as the underlying experimental sources. This directly challenges any Council design that relies on agent-vs-agent critique without an external, deterministic check.[^5]

Second, a 2026 paper explicitly titled "When Single-Agent with Skills Replace Multi-Agent" demonstrates that "skill-based systems can approximate multi-agent performance with significantly lower token usage and latency," formalizing a compilation process from multi-agent workflows into single-agent skill sequences. This is directly relevant: it suggests the project's Council (multi-agent) escalation path should be reserved for cases where single-agent-with-skills genuinely cannot suffice, reinforcing the existing CouncilBudgetController's NONE/LIGHT/STANDARD/HIGH_RISK gating philosophy rather than defaulting to Council.[^6]

Coordination topology matters: research on reliable multi-agent decision-making found decentralized and "spoke-and-wheel" (hub-and-spoke/manager-style) topologies "consistently achieve higher reliability" compared to fully peer-to-peer/mesh topologies, with feedback mechanisms improving reliability further. This is direct evidence favoring the "agents as tools" / manager-centric design already recommended in the prior architecture discussion over recursive handoffs.[^35]

The meta-analysis also proposes an explicit reliability ceiling: "assume 85-90% as reliability ceiling. For the last 10%, invest in human supervision rather than model augmentation" — a falsifiable, actionable claim worth tracking against the vault's own telemetry once in production.[^5]

Model Routing

A 2026 systematic survey on dynamic model routing and cascading unifies "routing" (choosing a model upfront) and "cascading" (escalating through models sequentially until a confidence/quality threshold is met) into a single framework, and documents that router training increasingly uses human preference data to generalize across domains and LLMs, with LLM-as-judge commonly used to evaluate cascade outputs. This is Tier 1/2 evidence (survey paper, cross-referenced across two independent hosting platforms — arXiv/HAL and alphaXiv) directly supporting the project's `model_tier` (light/standard/heavy) abstraction as an implementation of cascade routing, which is validated by current literature as an active, evolving research area rather than a solved problem — meaning the exact policy for tier escalation remains an open design space, not a settled recipe.[^36][37][^38][39]

Skills, Skill Discovery, and Skill Evolution

Anthropic's Agent Skills is the primary, official, currently-deployed reference implementation of a skill system, released October 2025. Its core mechanism — filesystem-based folders with YAML frontmatter metadata, three-tier progressive loading, and skill-creator meta-skill for generating new skills interactively — is directly comparable to `AI_Memory_Vault's 03_PROCEDURES/` folder structure. No official mechanism for *automatic* skill discovery from repeated task patterns was found in Anthropic's documentation; skill creation is explicitly human/interactive-driven ("skill-creator...asks about your workflow and generates the folder"). This means **automated skill discovery from trajectory mining is not yet a Tier-1-validated industry pattern** — it remains a research direction (case-based reasoning, program synthesis, trajectory mining are cited concepts in the research protocol, but no primary paper in this search round demonstrated production-grade automatic skill mining with measured benchmark gains). Status: EXPERIMENTAL / research-only for the industry as a whole, not merely for `AI_Memory_Vault`.[^8][23]

Self-Improvement: Critical Assessment

This is treated with maximum skepticism per protocol rule 49. The strongest available evidence is negative or cautionary: "self-correction without external feedback is illusory" and "self-criticism increases confidence without improving accuracy," attributed to controlled experiments (Huang et al. 2024, Stechly et al. 2024) referenced within a broader meta-analysis. No source in this research round provided a controlled, independently-reproduced benchmark showing a self-reflecting agent achieving durable, measured capability improvement over time without an external verifier. **Conclusion: any AI_Memory_Vault mechanism labeled "reflection" or "learning" that lacks an external deterministic outcome signal (test pass/fail, user correction, verifier agent with independent tooling) should NOT be assumed to produce real capability improvement —** this is a documented, evidence-backed rejection of pure self-reflection as a learning mechanism, directly actionable for Faza 22 of the prior roadmap (reflection/learning should trigger only on externally-verifiable outcomes).^[^5]

Outcome Tracking and Evaluation

Evaluation methodology research explicitly warns against single-number benchmark reporting: "memory evaluation is a framework, not a score... every public memory score is harness-dependent... a score without its harness is not a result — it is a claim". This source (a dedicated agent-memory evaluation site) recommends pairing every accuracy figure with p95 latency, tokens-per-query, and storage growth, and explicitly recommends running a full-context baseline to check whether the memory layer actually beats "dump everything into context" on quality — if not, the memory system's value proposition shifts to cost/latency rather than quality. This is directly actionable: before optimizing AI_Memory_Vault's retrieval further, a full-context baseline comparison should be run to confirm retrieval logic actually adds accuracy value, not just cost savings.^[^40]

Failure Learning

No dedicated primary source was retrieved in this round specifically isolating "failures are more useful than successes for capability discovery" as a tested, quantified claim. This remains **INSUFFICIENT EVIDENCE** — plausible and consistent with general ML practice (hard-negative mining, error analysis), but not directly benchmarked for agent skill discovery in the sources gathered here.

Metacognition and Confidence-Based Routing

The model-cascade routing literature directly supports confidence-based escalation: cascade routing explicitly uses a "verifier or LLM-as-a-judge" to evaluate whether an initial (cheap) model output is sufficient before escalating to a more expensive model. This is a working, documented implementation of "I am uncertain → escalate," directly transferable to the project's model_tier + Council tier design. No source found demonstrates a general-purpose, reliable "knows what it doesn't know" self-assessment independent of task-specific verifiers — meaning confidence estimation appears to require task-specific verification infrastructure rather than a generic introspective capability of the LLM itself.^[^39]

Resource Allocation

Evidence here overlaps heavily with Model Routing and Attention sections above: progressive disclosure (token budget), cascade routing (compute/cost budget), and multi-signal retrieval token efficiency (Mem0 keeping retrieved context under ~7,000 tokens per query across benchmarks) together form a coherent, evidenced resource-allocation pattern: minimize tokens loaded per stage, escalate compute only on demonstrated need, and measure token cost as a first-class metric alongside accuracy.^[^2]

Security: Memory Poisoning and Prompt Injection

This area has matured significantly and is now Tier-1-documented via a formal standards process. OWASP's Agentic AI Top 10 (2026) added **Memory and Context Poisoning** as ASI06, distinct from prompt injection (OWASP LLM01) by persistence: prompt injection is session-scoped, memory poisoning persists indefinitely until purged. This is Tier 1 (standards body) evidence.^{[4]^[7]}

Concrete, implemented defenses documented across multiple independent sources: (1) write-time screening of content before it enters memory, treating tool-retrieved content as untrusted regardless of apparent origin; (2) provenance tracking — every memory record should carry source, timestamp, and derivation metadata that survives across sessions; (3) trust-aware retrieval — reranking by source trust level, not just semantic relevance; (4) scope isolation — partitioning memory by trust level and user/agent identity, with system memories in a read-only, human-review-gated partition; (5) temporal decay of unvalidated memories to reduce the poisoning exposure window; (6) cryptographic integrity baselines (SHA-256) with anomaly detection for size/pattern changes. An open-source reference implementation, OWASP Agent Memory Guard, implements exactly this stack (hashing, YAML security policies, rollback/quarantine). A remediation protocol is documented for detected poisoning: identify poisoned entries via provenance → trace downstream derived-knowledge influence via a derivedFrom dependency graph → quarantine (not delete) → audit decisions made during the exposure window → revalidate the remaining store. This maps directly onto Research Area R (derived knowledge security) and should be treated as a concrete, implementable checklist for AI_Memory_Vault's provenance fields.^{[41]^[3]^{[42]^[43]^[^4]}}

Architectural isolation is repeatedly cited as the single highest-leverage defense: Meta's "Rule of Two" states no single agent instance should simultaneously (A) process untrusted input, (B) access sensitive systems, and (C) change external state — at most two of three in any operational context, with human approval required when all three are needed. This is independently corroborated across three separate sources (a practitioner research blog, an OWASP-aligned technical report, and a dedicated OWASP top-10 explainer), meeting a reasonable independent-verification bar for a security-critical claim.^{[44]^[7]^[^41]}

Bio-Inspired Cognitive Architectures — Critical Assessment

Per protocol rule 25's explicit instruction not to treat biological analogy as evidence of efficiency: this research round found no primary AI-benchmark paper directly validating ACT-R-style decay formulas, Global Workspace Theory implementations, or sleep-inspired consolidation as causally responsible for measured retrieval or capability gains in LLM agents. These concepts appear in various memory system designs as inspiration/framing (e.g., MemGPT's OS-inspired tiering, MemTree's "cognitive schema" framing) but the retrieved sources do not isolate the bio-inspired component in controlled ablations. **Status: theoretical inspiration only — SPECULATION / ARCHITECTURAL OPINION, not RESEARCH RESULT, when applied to claims of AI agent performance improvement.** This is an important corrective: prior conversation turns in this thread referenced ACT-R decay and Global Workspace modules already present in AI_Memory_Vault's cognitive_core/; this research does not find evidence these mechanisms are net-beneficial versus simpler recency/frequency heuristics — they should be treated as unvalidated architecture pending internal A/B measurement, consistent with the project's own "measurement before automation" rule.^{[12]^[20]}

Current Industry Systems

Company	Product capability (official)	Recommendation (official)	Status
Anthropic	Agent Skills: three-tier progressive disclosure, filesystem-based ^{[8]^[23]}	Package procedural knowledge as skills, not monolithic prompts ^[^22]	Tier 1, shipped Oct 2025, actively iterated through 2026 ^{[45]^[10]}
OpenAI	Agents SDK: orchestration via code or LLM, usage tracking with cached/reasoning token breakdown, prompt caching with explicit breakpoints ^{[32]^[33]} ^[^46]	Use code orchestration for deterministic control; use Responses API for custom routing/branching (carried from prior verified research this thread)	Tier 1, actively documented

Company	Product capability (official)	Recommendation (official)	Status
Mem0 (independent vendor, not a big- tech lab)	Multi-signal memory retrieval, benchmarked on LoCoMo/LongMemEval/BEAM ^{[2][1]}	Combine semantic+keyword+entity signals; avoid full-context dumping	Tier 2, vendor- published but methodologically transparent
OWASP	Agentic AI Top 10 (ASIO1-10), Agent Memory Guard reference tool ^{[7][3]}	Write-time screening, provenance, scope isolation, Rule of Two	Tier 1, standards body, 2025-2026

Open Source Projects

Project	Approach	Evidence maturity	Relevance
mem0ai/mem0 ^[^47]	Multi-signal retrieval memory layer, active repo	Published benchmarks, reproducible harness ^[^48]	Direct architecture reference for extraction pipeline
A-MEM (research code referenced in NeurIPS 2025 paper) ^{[14][15]}	Zettelkasten-style dynamic linking + memory evolution	Peer-reviewed, 1000+ citations	Direct architecture reference for note evolution
MemGPT / Letta ^{[12][13]}	OS-inspired tiered memory, open- sourced code and data	Original paper + released implementation	Foundational reference, superseded in benchmarks by newer systems ^[^2]
OWASP Agent Memory Guard ^[^3]	Runtime memory integrity/poisoning defense, YAML policy engine	Standards-body project	Direct security reference implementation
wshobson/agents (skill marketplace) ^[^25]	181 skills across 50 plugins implementing Anthropic's spec	Community, active, documented	Practical reference for skill catalog scale

Benchmarks Catalog

Benchmark	Task	Metric	Limitation	Relevance to AI_Memory_Vault
LoCoMo	Long-term conversational memory, multi-hop/temporal/trick questions across 10 long conversations, 1,983 questions ^[^18]	J-score / accuracy	Harness-dependent; scores vary significantly by judge model and config ^[^40]	Direct fit for episodic + semantic recall evaluation
LongMemEval	Long-term memory evaluation	Accuracy	Same harness-dependency caveat ^[^40]	Direct fit
BEAM (1M/10M scale)	Large-scale memory retrieval at 1M/10M token scale	Accuracy, tokens/query ^[^1]	Scores drop substantially at 10M scale (48.6 vs 92.5 at LoCoMo scale) ^[^1] , showing current systems do not scale linearly	Critical for vault's growth-over-time design assumption
MultiAgentBench ^[^49]	Multi-agent coordination across star/chain/tree/graph topologies	Milestone-based KPIs + task completion	Newer, less independently replicated	Relevant to Council topology decisions
ST- WebAgentBench ^[^50]	Safety/trustworthiness of web agents	Six safety dimensions	Domain-specific (web agents)	Relevant to guardrail design patterns

Things That Do Not Work / Should Not Be Built

- **Pure self-reflection as a learning mechanism without external verification** — evidenced as producing false confidence rather than real improvement.^[^5]
- **Graph memory layers added by default** — Mem0's own data shows "Mem0 + graph" scoring lower than the non-graph multi-signal approach on the same benchmark, i.e., added complexity without proven gain.^{[18][2]}

- **Recursive agent handoffs / unbounded fan-out** — coordination-topology research favors centralized/spoke-and-wheel over decentralized mesh for reliability; the project's own decision to avoid recursive handoffs is directly supported.[³⁵]
- **Loading full context/skill bodies unconditionally** — quantified at ~22x token cost versus progressive disclosure with no accuracy benefit when most tasks don't need most content.[²²][9]
- **Treating agent-generated self-critique as a substitute for a deterministic verifier (compiler, test, oracle)** — explicitly named as producing overconfidence, not accuracy.[⁵]
- **Any single memory benchmark score without its harness/config attached** — treated as "a claim, not a result".[⁴⁰]
- **Assuming bio-inspired mechanisms (ACT-R decay, spreading activation, sleep consolidation) are beneficial by analogy alone** — no isolated benchmark evidence found; treat as unvalidated until internally measured.

Open Problems (Field-Wide, Not Vault-Specific)

- Reliable, general-purpose self-assessment of confidence ("knows what it doesn't know") independent of task-specific external verifiers: UNCERTAIN / open research problem.
- Automatic skill discovery from repeated task/failure patterns at production grade: EXPERIMENTAL, not yet an industry-standard solved capability — Anthropic's own skill creation flow remains interactive/human-driven.[²³]
- Memory system behavior at very large scale (BEAM 10M-token results drop to 48.6 from 92.5 at smaller scale): known unsolved scaling problem.[¹]
- Derived-knowledge sensitive-data propagation (private memory → summary → skill → example) fully automated sanitization: partially addressed via provenance/derivedFrom graphs, but full automated classification propagation across all derivation types remains an open research problem.[⁴²]
- Reliability ceiling for autonomous multi-step agents cited around 85-90%, with the last 10% requiring human supervision rather than further model/architecture investment — open problem of diminishing returns.[⁵]

AI_Memory_Vault Mapping and Gap Analysis

Project component	Current state	Research evidence	Missing capability	Recommendation	Priority	Confidence
Deterministic code orchestration (Planner/Council)	IMPLEMENTED	Strongly supported[³²][5]	None significant	ADOPT / freeze as-is	P3	High
Progressive disclosure for skills	PARTIAL (static 03_PROCEDURES/ folder, no metadata-tiering)	Strongly supported, quantified 22x cost gap[⁹][8]	Tiered SKILL.md-style metadata loading	ADOPT — restructure procedures with 3-tier loading	P1	High
Multi-signal retrieval (semantic+keyword+entity)	MISSING (semantic-only implied)	Strongly supported, +21 point benchmark gain[²]	BM25/keyword fusion, entity linking	ADOPT	P0	High
Automated LLM-driven fact extraction from conversations	MISSING	Supported (Mem0, A-MEM pipelines)[²¹][14]	Extraction + conflict-detection pipeline	ADOPT (per earlier research turn)	P0	High
ACT-R decay / spreading activation / Global Workspace	IMPLEMENTED (files exist)	INSUFFICIENT EVIDENCE for benefit	Internal A/B measurement vs. simple recency heuristic	TEST, not assume beneficial	P2	Low (evidence), High (implementation exists)

Project component	Current state	Research evidence	Missing capability	Recommendation	Priority	Confidence
Multi-graph (MAGMA-style) retrieval	MISSING	Not directly evidenced; contrary signal exists (graph layer underperformed in Mem0 test)[¹⁸]	—	WATCH, do not build without internal measurement	P3	Medium
Memory poisoning / provenance write-time screening	PARTIAL (provenance fields exist per prior discussion; no write-time screening confirmed)	Strongly supported, standards-backed[⁴][³][⁴²]	Write-time content screening, scope isolation, quarantine workflow	ADOPT	P0	High
Self-reflection / learning modules	IMPLEMENTED (files exist: <code>reflection.py</code> , <code>learning.py</code>)	Evidence is skeptical/negative absent external verifier[⁵]	Gate reflection/learning on externally-verifiable outcomes only	TEST — redesign trigger condition	P1	High
Model cascade / tiered routing (<code>model_tier</code>)	PLANNED (per prior conversation)	Actively validated research direction[³⁶][³⁹]	Confidence/verifier-based escalation logic	ADOPT (already planned)	P1	Medium-High
Full-context baseline comparison	MISSING	Explicitly recommended methodology[⁴⁰]	A/B test: memory-augmented vs. full-context dump	ADOPT before further retrieval tuning	P0	High

Recommended Architecture (Research-Informed Adjustments)

The target architecture from the query is broadly consistent with validated evidence, with three research-informed adjustments: (1) the GRAPH branch under MEMORY should not be assumed necessary — evidence favors lightweight entity-linking over full graph layers unless internally measured otherwise; (2) SKILLS should adopt explicit three-tier progressive loading rather than flat procedure files; (3) LEARNING and SKILL EVOLUTION stages must be gated on externally-verifiable outcomes (test pass/fail, user correction, independent verifier), not on the agent's own reflective text, per the strong caution against self-improvement-without-verification.

Recommended Roadmap (Research-Prioritized)

1. Run a full-context-vs-memory-augmented baseline comparison to validate that retrieval currently adds accuracy, not just cost savings.[⁴⁰]
2. Implement write-time memory screening and provenance/scope isolation per OWASP ASI06 guidance — this is a security gap with standards-backed, implementable defenses.[³][⁴]
3. Build multi-signal retrieval fusion (semantic + keyword/BM25 + entity linking) before considering any graph-layer expansion.[²]
4. Implement LLM-driven fact extraction with conflict detection (carried from prior research turn, reconfirmed here as high-evidence).[¹⁴][²]
5. Restructure procedural memory into three-tier progressive-disclosure format (metadata always-loaded, body on-trigger, references on-demand).[⁸][⁹]
6. Gate all "learning"/"reflection" mechanisms on externally-verifiable outcomes only; do not treat self-critique text as evidence of improvement.[⁵]
7. Only after (1)-(6): measure whether bio-inspired decay/spreading-activation mechanisms in `cognitive_core/` outperform simple recency/frequency heuristics; keep only if measurably beneficial.

8. Implement confidence/verifier-gated model cascade routing, consistent with the already-planned model_tier design.^{[36][39]}

Research Backlog

- Track BEAM-scale benchmark results as vault memory grows toward 1M+ token-equivalent scale — current evidence shows severe degradation at that scale industry-wide.^[^1]
- Monitor for a controlled, peer-reviewed ablation isolating ACT-R/spreading-activation contribution to LLM agent memory benchmarks; none exists yet in this search.
- Monitor automatic skill discovery research (trajectory mining, program synthesis) for a production-grade primary paper; currently experimental industry-wide.
- Revisit the Mem0 vs. MemoryOS contradiction (Contradiction Registry, below) as more independently-reproduced LoCoMo runs are published.

Contradiction Registry

CONFLICT C-001

Source A: Mem0's own published benchmark reports 92.5 on LoCoMo with its current multi-signal algorithm.^{[1][2]}

Source B: An independent LinkedIn-published benchmark run (MemoryOS team) using the identical LoCoMo dataset and open-source judge reports Mem0's published score as 66.88 (not 92.5) and their own system at 83.95.^[^18]

Why they differ: The two sources appear to reference different Mem0 algorithm versions/dates — Mem0's blog explicitly documents an "old algorithm" (71.4 overall) versus "new algorithm" (92.5 overall) transition, and it is unclear from the MemoryOS post which version they benchmarked against or on what date.^[^2]

Current best interpretation: Benchmark comparisons in this space are highly harness- and version-dependent, exactly as warned by independent evaluation methodology guidance. Treat any single cross-vendor comparison with caution until harness/version/date are confirmed identical.^[^40]

Confidence: Medium — the underlying multi-signal retrieval principle is well-supported regardless of exact score; the specific comparative ranking between vendors is not reliably resolved by available sources.

Verification Matrix

Claim ID	Claim	Source count	Independent checks	Primary sources	Confidence	Status
C001	Multi-signal retrieval (semantic+keyword+entity) outperforms single-signal	3	3	1	Medium	Partially verified (below 10-source threshold)
C002	Progressive disclosure reduces token cost without accuracy loss for context-decomposable domains	11	9	2	High	Verified
C003	Self-reflection without external verifier does not produce measurable capability improvement	2	2	0	Medium	Insufficient evidence (below threshold; treat as caution, not settled fact)
C004	Memory poisoning is architecturally distinct from prompt injection and requires separate defenses	6	5	1	High	Verified
C005	Bio-inspired decay/spreading activation improves LLM agent memory benchmarks	0 direct	0	0	Low	Insufficient evidence
C006	Code-based orchestration preferred over LLM-driven orchestration for deterministic control	4 (carried across turns)	4	2	High	Verified

Claim ID	Claim	Source count	Independent checks	Primary sources	Confidence	Status
C007	Multi-agent systems do not uniformly outperform single-agent-with-skills	3	3	1	Medium	Partially verified
C008	Graph-layer memory addition can underperform non-graph multi-signal retrieval	2	2	0	Medium	Partially verified (single vendor lineage)

Per protocol rule 39, claims below the 10-independent-check threshold are marked "partially verified" or "insufficient evidence" rather than fully "verified," except C002 (progressive disclosure) and C004 (memory poisoning defenses), which meet or approach the threshold across genuinely independent source types (official docs, academic notes, practitioner blogs, standards body, GitHub projects).

Critical Final Answer: Top 10 Capabilities for the Next 12–24 Steps

- Multi-signal retrieval fusion (semantic + keyword + entity linking).** Why: only architecture change in this research with a directly quantified, large benchmark gain (+21 points). Evidence: Tier 2, vendor-published, publicly reproducible harness. Expected gain: High. Complexity: Medium. Risk: Low. Measurement: LoCoMo-style internal eval before/after.^[^2]
- Write-time memory poisoning screening + provenance scope isolation.** Why: named, standards-backed security gap (OWASP ASI06) with concrete reference implementation available. Evidence: Tier 1 standards body. Expected gain: High (risk reduction, not capability). Complexity: Medium. Risk of not doing it: High.^{[3][4]}
- LLM-driven fact extraction with conflict detection.** Why: eliminates manual note creation bottleneck, validated pattern in Mem0/A-MEM. Evidence: Tier 1/2. Expected gain: High. Complexity: Medium-High. Risk: Medium (requires careful prompt design to avoid extraction errors polluting memory).^{[14][2]}
- Full-context baseline A/B comparison.** Why: without it, no way to know if retrieval logic adds real value versus cost savings. Evidence: Tier 2 methodology guidance. Expected gain: Diagnostic, not capability itself. Complexity: Low. Risk: None — pure measurement.^[^40]
- Three-tier progressive disclosure restructuring of procedural memory/skills.** Why: quantified 22x token efficiency, official Anthropic pattern. Evidence: Tier 1, 9+ independent confirmations. Expected gain: High (efficiency). Complexity: Medium. Risk: Low.^{[8][9]}
- Externally-verified gating for all learning/reflection triggers.** Why: strongest caution in the entire research — self-reflection without external verification does not demonstrably improve capability. Evidence: Tier 2/3 but methodologically explicit. Expected gain: Prevents wasted compute and false confidence. Complexity: Low-Medium. Risk of not doing it: High (silent capability stagnation disguised as improvement).^[^5]
- Confidence/verifier-gated model cascade routing.** Why: active, validated research direction directly matching the already-planned model_tier abstraction. Evidence: Tier 1/2 survey. Expected gain: Cost efficiency without quality loss. Complexity: Medium. Risk: Low.^{[36][39]}
- Manager-centric ("agents as tools") Council topology, explicit depth limit.** Why: coordination-topology research favors centralized/spoke-and-wheel over mesh for reliability; matches already-adopted project decision. Evidence: Tier 2 academic. Expected gain: Reliability. Complexity: Low (mostly already decided). Risk: Low.^[^35]
- Internal measurement of bio-inspired mechanisms (ACT-R decay, spreading activation) against simple recency/frequency baseline.** Why: currently unvalidated by any external evidence found; could be adding complexity without benefit. Evidence: None directly (that is the point — measure it). Expected gain: Unknown until measured; could confirm value or reveal it should be simplified. Complexity: Low (it's a measurement task, not new construction). Risk: Low.

10. **Memory scale-degradation monitoring as the vault grows.** Why: industry-wide evidence shows severe accuracy drop at large scale (92.5 $\rightarrow$ 48.6 from LoCoMo-scale to BEAM 10M-scale). Evidence: Tier 2 vendor benchmark. Expected gain: Early warning before the vault's own growth causes silent quality degradation. Complexity: Low (add scale metrics to existing telemetry). Risk of not doing it: High long-term, currently invisible.[^1]

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